

Active v0.3.0 - frozen at b8d8d93; final metrics and publication pending

Task: real-versus-AI classification with a label, AI-positive score, CPU web/API/batch interface and model-influence explanation. Public-data results only; organizer data, baseline and hidden scores were not supplied.

Model, data and operating point

Frozen generic **CLIP ViT-L/14** image tower plus our trained logistic head. Training: **8,000 CIFAKE + 6,239 GenImage BigGAN/SD1.5** images; original and identical crop/JPEG views, 90% GenImage loss weight. C=10 selected on internal validation. The pretrained tower is credited to OpenAI; we trained the head.

Inference: EXIF orientation, official 224px CLIP centre crop, L2-normalized embedding, linear logit and temperature scaling. Temperature **0.946907**; AI threshold **0.821428**. Temperature uses separate calibration data; threshold uses internal validation, not external labels.

Calibration: 9,969 CIFAKE + 781 GenImage; validation sizes below. Exact/perceptual overlap screening and grouped splits are heuristic. Backbone pretraining overlap is unknown. GenImage-derived head: noncommercial research use.

Runnable interface and attribution

Shared CPU CLI/API; single image or batch with one loaded model. JPEG/PNG/WebP uploads up to **25 MiB / 40 MP**. EXIF is separate and does not alter predictions; C2PA is not checked.

Tower **608,352,029 bytes** + head **7,949 bytes**, SHA-256 verified. **CPU-only** model runtime: torch, without clip/torchvision/ftfy/regex; ordinary app dependencies remain required. Public assets and clean-download setup timing are pending.

Measured CPU components: predict **~0.38s/image** (ResNet ~64ms), explanation **~5.5s**, robustness **~5.1s**. Not combined-request guarantees. Input-gradient influence map uses a 14px patch grid plus masking diagnostic.

Measured development performance at the stated operating point

Evaluation	AUC	Macro-F1	Accuracy	Real FPR	AI recall
GenImage validation (783)	0.991	0.950	95.0%	1.9%	92.3%
CIFAKE validation subset (4,000)	0.910	0.759	76.7%	5.0%	58.4%
External dev - original	0.771	0.604	63.5%	11.7%	38.8%
External dev - format matched	0.789	0.608	64.2%	10.0%	38.5%

External rows average two generator/source pairs, 2,000 unique dev images per format. AUC is ranking, not accuracy. Training-encoder metrics; CPU/export checks found zero label disagreements on measured samples with score drift up to 0.0097. GenImage validation confusion matrix (rows actual real/AI, columns predicted real/AI): **[[362, 7], [32, 382]]**.

Baseline and FAILED gate: v0.2.0 to v0.3.0 mean AUC **0.647 to 0.771** original, **0.653 to 0.789** matched; guided matched **0.611 to 0.808**, LDM matched **0.695 to 0.769**. But **as_distributed_idm_200_fpr FAILED**: real FPR **22.0% vs 12.8%** (allowed +5 points). Original, stricter-threshold and COCO attempts all failed this check. Selected model accepts this tradeoff. Guided recall at a diagnostic 5% FPR budget: **13.0% to 42.6%**; externally fitted comparison thresholds are not deployed.

Final evaluation - pending

[PENDING - reserved evaluation in progress, will be filled before submission]

Overall/per-generator AUC, macro-F1, accuracy, FPR and confusion matrices: GLIDE/DALLE, original/matched. Disclosed **second use** of the public reserve after freeze, not a fresh blind test; selection used development data. COCO's 354 reserved reals: first-use FPR with Wilson 95% intervals, not AI accuracy. No retuning. v0.2.0 archives preserved.

User diagnostic and reference

Private 11 ChatGPT + 18 phone images, v0.2.0 to CLIP: as-uploaded AI detections **0/11 to 9/11**, real flags **8/18 to 9/18**; matched detections **2/11 to 9/11**, real flags **2/18 to 4/18**. Small-sample CLIP AUC 0.732/0.788. **No accuracy percentage claim from 29 images**; no fitting, per-image detail private.

B-Free reference: 0.970 original / 0.945 matched development AUC on the same images. Third-party weights, not our trained result or deployed component.

Limitations and submission checks

Weaker explanation localisation: only **17/40** audit images satisfy the masking-support rule; the other 23 are disclosed as not localised. Deletion tests p=0.29/0.82. No verified visual-defect or AI-object localisation; human usefulness review pending.

Real-photo false positives remain high; unknown generators/calibration can fail and centre crops omit borders. New-model degradation benchmarks and updated demo pending; screenshot simulation is not device capture. No generator attribution, caption consistency or C2PA verification.